Visual Interfaces for Understanding and Shaping Complex Information With AI

From the stories and videos we encounter every day to the datasets and models used in scientific research, the world is full of complex information whose underlying structure can be challenging to access directly. For example, a machine learning (ML) researcher may want to uncover patterns a transformer model has learned, while a video creator may want to see how their footage fits together into a story and where key moments are missing.

My research explores how **artificial intelligence (AI) and visualization can work together to make this hidden structure visible and actionable**. I develop *AI-mediated visual interfaces* that expose or construct meaningful representations of complex information, then transform them into interactive views to support new forms of understanding and creation. Across these systems, AI serves three complementary roles: an object of analysis, helping reveal otherwise hidden patterns and relationships inside models; a structuring mechanism, deriving meaningful organization from unstructured content; and a collaborator in creation, allowing users to reshape and augment information through visual structure (Figure 1).

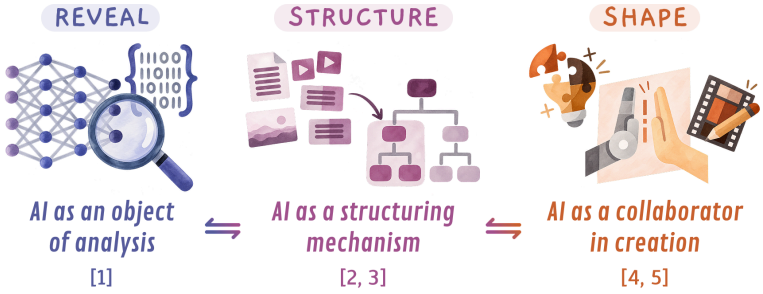


Figure 1: My research develops AI-mediated visual interfaces in which AI serves as an object of analysis, a structuring mechanism, or a collaborator in creation. Together, these interfaces support new forms of understanding and shaping complex information.

My work lies at the intersection of **human-computer interaction (HCI), data visualization, and artificial intelligence**. It has resulted in five first-author publications at top HCI and visualization venues including ACM CHI and IEEE VIS, with **three receiving Best Paper Honorable Mention** awards. I am also supported by the **NSF Graduate Student Fellowship** and **Kempner Institute Graduate Fellowship.** Beyond these recognitions, my collaborations with industry labs have catalyzed follow-up research and new product directions at Google DeepMind, Adobe, Apple, and Microsoft. I have designed and evaluated systems with users ranging from ML researchers and professional red teamers to literary scholars and video creators, grounding my research in real-world analytical and creative practices.

🔍 *Revealing Structure Within AI Models*

Modern ML models learn rich internal representations that shape their behavior, but these representations are difficult to interpret due to their high-dimensional, distributed nature. Visualization can help expose this latent organization, but existing approaches often focus on individual examples or isolated representation spaces. My research develops interactive visualization techniques that reveal relationships across inputs and levels of detail, enabling richer analyses of model behavior.

AttentionViz: A Global View of Transformer Attention [1]

Transformer models use self-attention to learn contextual relationships between sequence elements (e.g., words in a sentence), making attention a useful lens for understanding model behavior. To make attention mechanisms more interpretable, I developed a technique that visualizes the *joint embedding of query and key vectors*, two core elements of attention. Unlike prior visualization techniques, this approach facilitates the exploration of *global patterns across many inputs*, providing a novel, higher-level view of attention. Building on this technique, I created AttentionViz, an interactive tool that visualizes attention at multiple levels in both language and vision transformers. Alongside ML researchers and

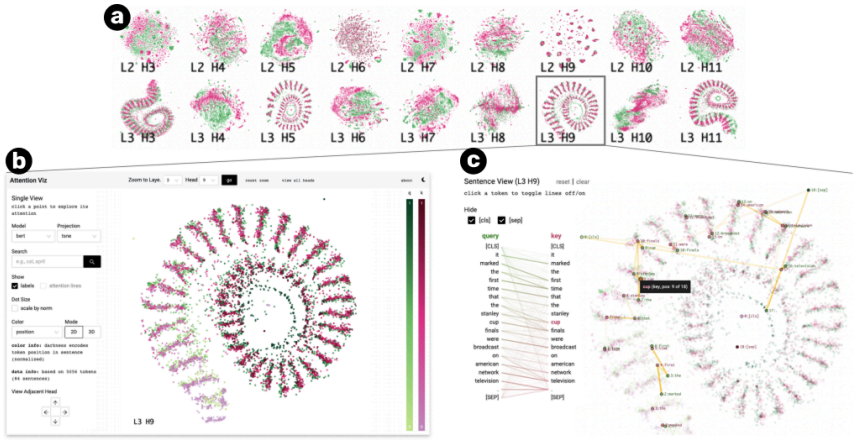


Figure 2: AttentionViz visualizes joint query-key embeddings in transformers to enable attention exploration at multiple levels: from a (a) global view of all attention heads to (b) individual heads and (c) input-level patterns. This "spiral" shaped head in BERT encodes a positional attention pattern.

vision scientists, we uncovered several insights with AttentionViz, including striking position-based attention patterns in language models (Figure 2) and hue- and frequency-based patterns in vision models, showing the utility of our approach in improving model understanding through interpretable structure.

Structuring Unstructured Information With AI

While AttentionViz reveals what models have learned, AI can also help uncover meaningful structure in other complex information. Stories and videos, for instance, contain characters, themes, and events whose relationships typically require significant manual effort to extract and visualize. My next line of work explores how generative models can derive task-specific structure from unstructured media, and how to design useful visualizations and interactions around these AI-extracted representations.

Story Ribbons: Extracting Narrative Structure With Large Language Models [2]

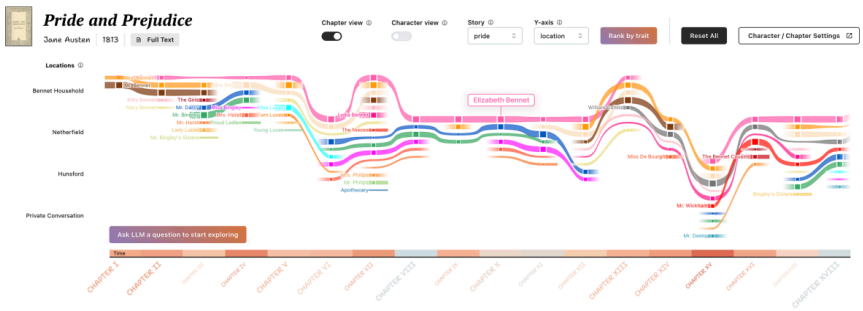


Figure 3: Story Ribbons visualizes LLM-extracted insights about literary works (e.g., Pride and Prejudice). Each "ribbon" represents a different character and tracks interactions across novel chapters (x-axis) and locations (y-axis).

Literary analysis involves tracking how characters, locations, and themes evolve across a story. Storyline visualizations make these interactions easier to explore, but producing the structured data needed to create one typically requires substantial manual annotation. To address this bottleneck, I developed a data pipeline that automatically *extracts relevant narrative information from novels and scripts with large language models (LLMs)*, using "correction loops" to detect hallucinated quotes and duplicate entities. Applying this pipeline, I created Story Ribbons, an interactive system co-designed with literary scholars for exploring character and theme trajectories across multiple narrative scales (Figure 3). Users can also define new dimensions (e.g., rank characters by "greed") and inspect explanations for model-generated judgments. Through pipeline and user evaluations with Story Ribbons on 36 literary works, we demonstrated how LLMs can streamline narrative visualization creation and enable new forms of literary analysis, while *highlighting interpretability as a key requirement for AI-assisted sensemaking.*

VideoTrail: Navigating Videos Through Adaptive Hierarchies [3]

Much like stories, videos contain rich semantic structure, yet most interfaces organize them around linear playback, making it difficult to find and revisit specific moments. Drawing on information foraging

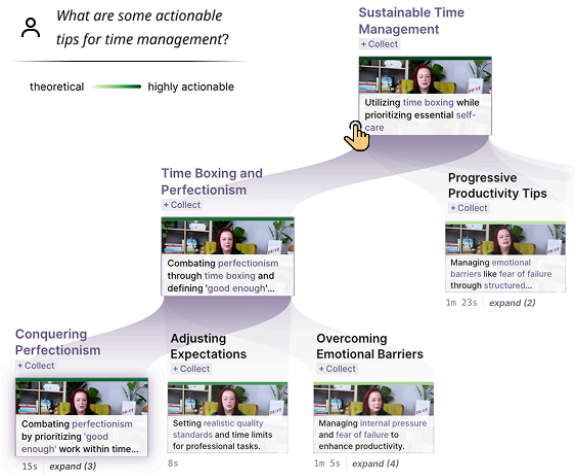


Figure 4: VideoTrail generates adaptive, hierarchical visualizations of videos to help users forage for information (e.g., actionable time management tips). Expanding a video segment reveals additional, finer-grained layers of the hierarchy.

theory, I developed VideoTrail, an AI-powered browser extension that uses a multimodal LLM to transform long-form videos into hierarchical, explorable roadmaps (Figure 4). VideoTrail lets users move between high-level overviews and detailed segments while surfacing *adaptive information scent* that prioritizes visual (e.g., keyframes) or textual cues (e.g., summaries) based on video content. In a user study, VideoTrail supported efficient, accurate information seeking through hierarchical search and enabled new foraging strategies for both unfamiliar and familiar videos, highlighting how AI-derived structure can augment existing workflows and make linear media more semantically navigable.

Shaping Information Through Visual Structure

Beyond helping people understand and navigate information, visual representations can also reveal opportunities for extending or reshaping it. However, for tasks such as data augmentation and video storytelling, a key challenge is identifying meaningful gaps and determining how to fill them in contextually relevant ways. My research contributes visual interfaces that make these gaps visible and enable users to steer AI-assisted generation to address them.

Amplio: Exploring Empty Spaces for Data Augmentation [4]

Robust and safe ML systems depend on diverse evaluation data, yet identifying gaps and determining where additional coverage is needed can be difficult and time-consuming. To address this challenge, I created Amplio, an interactive tool that visualizes unstructured text as a semantic embedding space to *surface underexplored regions and guide targeted data augmentation*. Users can fill these gaps through three human-in-the-loop techniques: *steering with interpretable concepts* learned by a sparse autoencoder, *interpolating between examples*, and *contextualized prompting* with LLMs (Figure 5). In a study with professional red teamers, Amplio helped participants quickly generate diverse, relevant examples to augment a model safety dataset, illustrating how visualization can support a tighter loop between understanding data coverage and actively reshaping it.

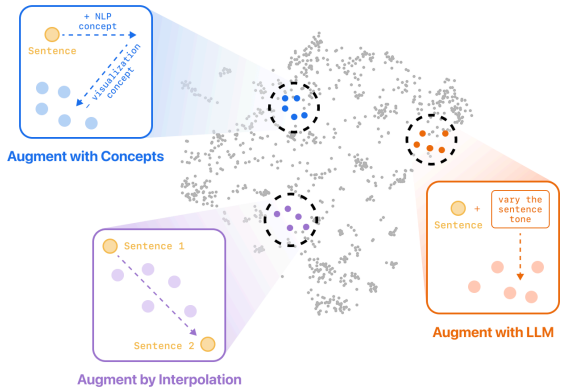


Figure 5: Amplio surfaces empty spaces in text datasets and includes three human-in-the-loop augmentation methods for filling in these regions to improve data diversity: Augment with Concepts, Augment by Interpolation, and Augment with LLM.

Vidmento: Expanding Video Stories Through Context-Aware Generation [5]

Similar challenges arise in video storytelling, where limited footage can constrain creative expression and leave undesired narrative gaps. To address this limitation, I developed Vidmento, a video authoring system that analyzes existing footage to identify opportunities for story development and then uses

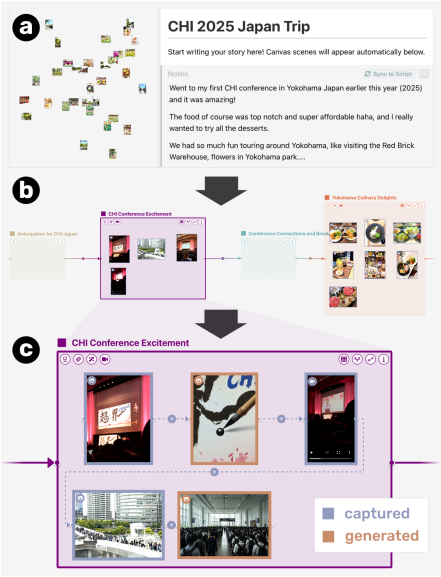


Figure 6: Shaping a video story about a trip to Japan with Vidmento from (a) an initial set of unstructured materials by (b) grouping shots into scenes and connecting them into a visual storyline. (c) Scenes can be expanded by sequencing existing shots and generating new connective moments.

generative video to create material that fits the surrounding narrative and visual context. Vidmento's interface acts as a *visual storyline* that users can reshape at the story, scene, and shot levels (Figure 6). In a study with video creators, Vidmento supported narrative development and exploration through its *context-aware expansion pipeline*, enabling expressive storytelling while preserving creative control over how generated media was incorporated. Together, Amplio and Vidmento demonstrate how visual representations can bridge understanding and creation by *making gaps in semantic structure visible, then giving users agency over how — and whether — to fill them*.

Future Research Agenda

My long-term goal is to design visual interfaces that help people **understand, steer, and create with increasingly capable AI systems**. I envision these representations as a shared medium between human intent and AI-derived structure, supporting sensemaking at scale, calibrated trust, and expressive control over complex information.

Scaling From Single Artifacts to Semantic Collections

My current systems primarily help people understand and shape individual datasets, stories, and videos. Foundation models create new opportunities to derive comparable semantic structure across entire collections, enabling visual sensemaking at a much larger scale. Building on the multiscale visualization approach in AttentionViz [1], I want to develop AI-mediated interfaces for exploring global patterns through careful structural alignment and aggregation across different artifacts. This could help literary scholars uncover recurring character dynamics or narrative structures across hundreds of novels, or ML researchers identify systemic failure patterns across model outputs and benchmarks. Ultimately, I aim to *transform large collections of unstructured content into interactive semantic spaces for comparison, discovery, and augmentation*.

Calibrating Trust in AI-Derived Structure

As interactive systems increasingly rely on structure inferred by generative models, they risk presenting subjective interpretations as ground truth. The way an LLM segments a video, identifies a theme, or rates character importance may vary across runs, prompts, and models, yet this variability is rarely communicated to users. Importantly, receiving different readings can also be meaningful, particularly in interpretive domains like literary analysis. Extending Story Ribbons [2] and VideoTrail [3], I aim to create interfaces that surface alternative interpretations, reveal sensitivity to model and prompting choices, and connect generated data back to source evidence. I am especially interested in *new visual encodings that communicate the reliability and stability of AI-derived structure,* helping people reason with these representations while also interrogating their limitations and calibrating trust in what they see.

Visualizations as Interfaces for Generative Steering

As Amplio [4] and Vidmento [5] show, many real-world tasks require understanding an artifact's existing structure before deciding how it should change. Building on this work, I want to develop *bidirectional*

visual representations that support both analysis and generative control. While current generative interfaces largely center on prompts and outputs, steerable visualizations could help users directly manipulate an artifact's semantic structure, explore alternatives, and inspect downstream effects. A key challenge is keeping the visualization and evolving artifact faithfully synchronized. For example, a writer could reshape a character's trajectory [2] and immediately see how that affects other narrative elements, while direct edits to the story would update the visualization. Persistent visual representations could similarly support agentic workflows such as code generation by helping users express intent and track alignment with their goals over long-horizon tasks.

References

[1] AttentionViz: A Global View of Transformer Attention. **Catherine Yeh,** Yida Chen, Aoyu Wu, Cynthia Chen, Fernanda Viégas, Martin Wattenberg. *IEEE VIS 2023*.

[2] Story Ribbons: Reimagining Storyline Visualizations With Large Language Models. **Catherine Yeh,** Tara Menon, Robin Singh Arya, Helen He, Moira Weigel, Fernanda Viégas, Martin Wattenberg. *IEEE VIS 2025*. 🏆 Best Paper Honorable Mention.

[3] VideoTrail: Interactive Video Foraging Through Hierarchical Visualization. **Catherine Yeh,** Bryan Wang, Fernanda Viégas, Martin Wattenberg. *In Submission*.

[4] Exploring Empty Spaces: Human-in-the-Loop Data Augmentation. **Catherine Yeh,** Donghao Ren, Yannick Assogba, Dominik Moritz, Fred Hohman. *ACM CHI 2025*. 🏆 Best Paper Honorable Mention.

[5] Vidmento: Creating Video Stories Through Context-Aware Expansion With Generative Video. **Catherine Yeh,** Anh Truong, Mira Dontcheva, Bryan Wang. *ACM CHI 2026*.

See my CV for a complete list of publications.